DAY 08 • STUDENT ACTIVITIES

Manual Checking and Verification

Checking detections and actions yourself, patching security solution updates, and scanning infected systems

Qualification	Cyber Threat Monitoring Level I
Unit of Competency	CS-ICT251101 Monitor and Report Cyber Threats
Coverage	Element 3 Perform manual checking and verification: Performance Criteria 3.1–3.4; Range of Variables: Action; Required Knowledge 3.1 Log and detection management, 3.2 Security solution scan and operations procedure, 3.3 Knowledge in organization SOP; Required Skill 3.6 Verification skills
Duration	8 hours (1 training day)

Trainee Name	Batch / Section	Date

Activities Overview

No.	Activity	Type	Time	Assets Needed
8.1	Verify the Detection	Individual (hands-on)	60 min	detections_to_verify.csv, samples/, threat_intel_hashes.txt, known_good_hashes.txt (lab: ~/day8)
8.2	Check the Action	Individual (hands-on)	45 min	remediation_events.csv, post_action_check.csv (lab: ~/day8)
8.3	Check and Patch the Updates	Individual (hands-on)	45 min	vendor_bulletin.txt, update_status.csv (lab: ~/day8); your WKS-LAB workstation
8.4	Scan the Infected System	Individual (hands-on)	60 min	CTM Virtual Lab: your WKS-LAB workstation (Update-MpSignature, Start-MpScan)

CAUTION — Lab Rules

- Work only on the lab machines, virtual machines and files provided by your trainer.
- Never scan, probe or test any system outside the authorized training network.
- Treat all sample logs and data as confidential, as if they belonged to a real client.
- The samples are harmless text files. Never download, open or run real malware samples.

Activity 8.1 — Verify the Detection

Type	Individual (hands-on)
Time Allotment	60 min
Objective	Check five detections yourself with their SHA-256 hashes (PC 3.1).
Assets / Materials	assets/detections_to_verify.csv; assets/samples/; assets/threat_intel_hashes.txt; assets/known_good_hashes.txt (CTM Virtual Lab: ~/day8/)

Instructions

1. Read rule V1: grep -A12 "^V1 CHECK" ~/day8/soc_verification_sop.txt
2. Read the console report: cut -d, -f1,2,3,6 ~/day8/detections_to_verify.csv
3. Hash every sample: sha256sum ~/day8/samples/* (or Get-FileHash on Windows).
4. Compare each hash with console_sha256, then grep it in threat_intel_hashes.txt and known_good_hashes.txt.
5. Choose the verdict: verified malicious, false positive, hash mismatch, or unknown (escalate).

Work Space

Detection	Hash = Console?	In Threat Intel?	Verdict	Notes
V-01				
V-02				
V-03				
V-04				
V-05				

Sample that changed after detection (meaning)	
False positive and its proof	

Deliverable

Completed detection verification (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Computed and compared every hash	☐	☐
Used both reference lists correctly	☐	☐
Recognized the changed file and escalated the unknown one	☐	☐

Activity 8.2 — Check the Action

Type	Individual (hands-on)
Time Allotment	45 min
Objective	Check the action of the security solution in its event log and on the host (PC 3.2).
Assets / Materials	assets/remediation_events.csv; assets/post_action_check.csv (CTM Virtual Lab: ~/day8/)

Instructions

1. Read rule V2: grep -A10 "^V2 CHECK" ~/day8/soc_verification_sop.txt
2. Show the action events: Import-Csv day8\remediation_events.csv | Format-Table event_id, host, action, detail
3. Read the after-action check: cat ~/day8/post_action_check.csv
4. For each host: action verified, not verified, or failed.
5. Choose the next step: record, or escalate to Tier 2.

Work Space

Host	Event ID	File Still Present?	Result	Next Step
HR-LT-07				
QA-PC-05				
LOAN-PC-08				
IT-PC-01				
PAY-PC-02				
DEV-PC-02				

Why IT-PC-01 is not verified	

Deliverable

Completed action verification (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Read the event ID for every action	☐	☐
Checked the host result, not only the event	☐	☐
Escalated the failed and unverified actions	☐	☐

Activity 8.3 — Check and Patch the Updates

Type	Individual (hands-on)
Time Allotment	45 min
Objective	Check the security solution versions against the vendor bulletin, patch your own workstation and verify it (PC 3.3).
Assets / Materials	assets/vendor_bulletin.txt; assets/update_status.csv (CTM Virtual Lab: ~/day8/); your WKS-LAB workstation

Instructions

1. Read the bulletin: cat ~/day8/vendor_bulletin.txt
2. Show the fleet versions: Import-Csv day8\update_status.csv | Format-Table
3. Mark each host up to date or out of date (signatures, engine or platform) and choose the action.
4. On your workstation: Get-MpComputerStatus | Format-List AntivirusSignatureVersion, AMEngineVersion, AMProductVersion, then Update-MpSignature.
5. Verify: the new version, and event 2000 in the Defender Operational log.

Work Space

Host	Signatures	Engine / Platform	Status	Action
DC-01				
FS-01				
PAY-PC-02				
FIN-PC-03				
HR-LT-07				
QA-PC-05				
LOAN-PC-08				
IT-PC-01				

My signature version before	
My signature version after	
Event ID that records the update	

Deliverable

Completed update check and verified patch (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Compared all three versions with the bulletin	☐	☐
Updated signatures and routed engine/platform updates to IT	☐	☐
Verified the patch with the version and event 2000	☐	☐

Activity 8.4 — Scan the Infected System

Type	Individual (hands-on)
Time Allotment	60 min
Objective	Use the security solution to scan your infected workstation following SOC-SOP-04 V4, and verify the result (PC 3.4).
Assets / Materials	CTM Virtual Lab: your WKS-LAB workstation (PowerShell)

Instructions

1. Update first: Update-MpSignature (if you did not in Activity 8.3).
2. Full scan: Start-MpScan -ScanType FullScan
3. What was found: Get-MpThreatDetection | Format-List Resources, ActionSuccess, ThreatStatusID
4. Events: Get-WinEvent for Ids 1001, 1116, 1117, 1118 on the Defender Operational log. Then Test-Path "<file path>".
5. Record the result. Every seat is different.

Work Space

Threat name	
File path	
Scan type used	
Action result (event ID)	
File still present?	
Next step	

Deliverable

Completed scan record (worksheet in your CTM Virtual Lab notebook).

Performance Criteria (Trainer Use)

Criteria	Yes	No
Updated before scanning and used a full scan	☐	☐
Verified the action with events and Test-Path	☐	☐
Correct next step for the result	☐	☐

Self-Check

A. Multiple Choice

Encircle the letter of the best answer.

1. The hash of a file on disk is different from the hash the console reported. This means:
    a) The detection is verified
    b) The file changed after the detection
    c) The file is a false positive
    d) The console is broken
2. Defender event 1118 means:
    a) Malware detected
    b) Action taken
    c) Action failed (non-critical)
    d) Signatures updated
3. Event 1117 says quarantined, but Test-Path returns True. The action is:
    a) Verified
    b) Not verified
    c) Routine
    d) A false positive
4. The signatures are older than the vendor bulletin. You should:
    a) Ignore it
    b) Run Update-MpSignature and verify
    c) Re-image the PC
    d) Uninstall Defender
5. On an infected system the SOP requires:
    a) A quick scan with old signatures
    b) An update, then a full scan
    c) No scan
    d) Deleting files by hand

B. True or False

Write TRUE if the statement is correct, FALSE if it is not.

No.	Statement	Answer
1	Computing a file's hash runs the file.	
2	"No threats found" after a quick scan proves the computer is clean.	
3	A hash found in the vendor's known-good list points to a false positive.	
4	Engine and platform updates are requested from IT.	

C. Short Answer

1. Why must you check the host (Test-Path) and not only the action event?

2. Why update the signatures BEFORE scanning an infected computer?

Reflection

What was the most important thing you learned today, and how will you apply it in a Security Operations Center?

Trainer Sign-off

Overall Result	Trainer Remarks	Trainer Signature / Date
☐ Competent		
☐ Not Yet Competent		